

Lecture 2

Naive definition of the probability and Counting

16.02.2026

Outline

- 1 Naive Definition of Probability
- 2 The basic rule of counting
- 3 Permutations
- 4 Combinations
- 5 Multinomial coefficients

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Naive Definition of Probability

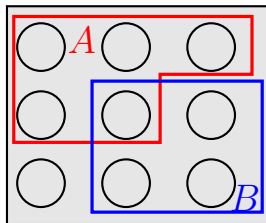
Definition

Let A be an event for an experiment with a finite sample space S . The naive probability of A is

$$P_{naive}(A) = \frac{|A|}{|S|} = \frac{\text{number of outcomes favorable to } A}{\text{number of outcomes favorable to } S}$$

- The naive definition is very restrictive in that it requires, S to be finite, equal mass to each outcome.
- It has often been misapplied by people who assume equally likely outcomes without justification and make arguments to the effect of “either it will happen or it won’t, and we don’t know which, so it’s 50-50”.

Example



Choosing a circle from the box shown in the figure.

A :

B :

$$P(A)$$

$$P(B)$$

$$P(A \cup B)$$

$$P(A \cap B)$$

$$P(A^c)$$

$$P(B^c)$$

$$P([A \cup B]^c)$$

$$P([A \cap B]^c)$$

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The basic rule of counting

- Suppose an experiment consists outcome 1 and outcome 2. If there are n_1 possibilities for outcome 1, and n_2 possibilities for outcome 2, then together there are

$$n_1 \times n_2$$

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Example: A team of one boy and one girl is to be made form a group of 5 girls and 2 boys. How many different teams are there? (Team A is different from Team B if at least one player is different.)

The generalized rule of counting

- Suppose an experiment consists r different outcomes, with the i -th outcome having n_i possibilities, then together there are

$$n_1 \times n_2 \times \cdots \times n_r = \prod_{i=1}^r n_i$$

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Example: how many different license plates?

_____	_____	_____	_____	_____	_____
letter	letter	letter	number	number	number

$$26 \times 26 \times 26 \times 10 \times 10 \times 10 = 17,576,000$$

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- Each of these arrangements is a permutation
- The order matters!
- Number of permutations of n different objects

$$n \times (n - 1) \times \cdots 2 \times 1 = n!$$

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Permutation of r groups of n objects

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Notice that there are two “o” in the word “Facebook”.

c	e	F	a	b	o_1	o_2	k
c	e	F	a	b	o_2	o_1	k

The same words! Different orders between the two “o” do not matter.

$$\frac{\#(\text{permutations if all different})}{\#(\text{permutations among the 2 “o”})} = \frac{8!}{2!}$$

Permutation of r groups of n objects

- Among n objects, if n_1 are alike, n_2 are alike, $\dots$, n_r are alike, then there are

$$\frac{n!}{n_1!n_2!\cdots n_r!}$$

different permutations.

Permutation of selecting r items from n objects

How many permutations of 2 numbers among $\{1, 2, 3, 4, 5, 6\}$ are there?

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If we have n items and want to select r of them,

$$\#(\text{permutations}) = n \times (n - 1) \times \cdots \times (n - r + 1) = \frac{n!}{(n - r)!}$$

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$$\binom{n}{r} = \frac{n!}{(n-r)!r!}$$

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- $0! = 1$

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Same questions: how many combinations of 2 numbers among $\{1, 2, 3, 4, 5, 6\}$ are there?

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$$\begin{array}{ccccccccc} \{1,2\} & \{1,3\} & \{1,4\} & \{1,5\} & \{1,6\} & & & & \\ & \{2,3\} & \{2,4\} & \{2,5\} & \{2,6\} & & & & \\ & & \{3,4\} & \{3,5\} & \{3,6\} & & & & \\ & & & \{4,5\} & \{4,6\} & & & & \\ & & & & \{5,6\} & & & & \end{array}$$

$$\binom{6}{2} = \frac{6!}{4!2!} = \frac{6 \times 5}{2 \times 1} = 15$$

Example: Poker hand. A standard poker deck has 52 cards, in four suits (clubs, diamonds, hearts, spades) of thirteen cards each (2, 3, ..., 10, Jack, Queen, King, Ace).

How many poker hands (5 cards) can be dealt from a deck of 52 cards?

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Combinations, choose 5 out of 52 different cards.

$$\binom{52}{5} = \frac{52!}{47!5!} = \frac{52 \times 51 \times 50 \times 49 \times 48}{5 \times 4 \times 3 \times 2 \times 1} = 2,598,960$$

Properties of combinations $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

1

$$\binom{n}{1} = n$$

$$\binom{n}{n} = 1$$

Properties of combinations $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

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2

$$\binom{n}{r} = \binom{n}{n-r}$$

Properties of combinations $\binom{n}{r} = \frac{n!}{(n-r)!r!}$

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$$\binom{n}{1} = n \qquad \binom{n}{n} = 1$$

2

$$\binom{n}{r} = \binom{n}{n-r}$$

3

$$\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}, \quad 1 \leq r \leq n$$

Binomial theorem

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

Proof: 1) mathematical induction or 2) combinatorial consideration.

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$$\binom{10}{5} \binom{5}{2} \binom{3}{3} = \frac{10!}{5!(10-5)!} \cdot \frac{5!}{3!(5-3)!} = \frac{10!}{5!3!2!}$$

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Multinomial coefficient: a set of n distinct items is to be divided into r distinct groups of respective sizes $n_1, \dots, n_r$, where $n_1 + n_2 + \dots + n_r = n$. Number of possible divisions is

$$\binom{n}{n_1, n_2, \dots, n_r} \stackrel{\text{def}}{=} \frac{n!}{n_1! n_2! \dots n_r!}$$



$$\binom{n}{n_1, n_2, \dots, n_r} = \binom{n}{n_1} \binom{n - n_1}{n_2} \binom{n - n_1 - n_2}{n_3} \dots \binom{n_r}{n_r}$$



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- When $r = 2$, becomes binomial coefficient (choose function)

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Note that $n_1 + n_2 = n$



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- Multinomial Theorem

$$(a_1 + a_2 + \dots + a_r)^n = \sum_{n_1 + \dots + n_r = n} \binom{n}{n_1, n_2, \dots, n_r} a_1^{n_1} a_2^{n_2} \dots a_r^{n_r}$$



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- The Binomial theorem is a special case when $r = 2$.

Counting examples

Among 4 married couples, we want to select a group of 3 people that is not allowed to contain a married couple. How many choices are there?

Counting examples

How many subsets are there of the set $1, 2, \dots, n$

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To fulfill the requirements for a certain degree, a student can choose to take any 7 out of a list of 20 courses, with the constraint that at least 1 of the 7 courses must be a statistics course. Suppose that 5 of the 20 courses are statistics courses. How many choices are there for which 7 courses to take?

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Explain why the answer is not $\binom{5}{1} \binom{19}{6}$?

Probability Naive definition examples

There are k people in a room. Assume each person's birthday is equally likely to be any of the 365 days of the year (we exclude February 29), and that people's birthdays are independent (we will define independence formally later, but intuitively it means that knowing some people's birthdays gives us no information about other people's birthdays; this would not hold if, e.g., we knew that two of the people were twins). What is the probability that at least one pair of people in the group have the same birthday?

Probability Naive definition examples

A 5-card hand is dealt from a standard, well-shuffled 52-card deck. The hand is called a full house in poker if it consists of three cards of some rank and two cards of another rank, e.g., three 7's and two 10's (in any order). What is the probability of a full house?